

Toward Automatic Building Footprint Delineation From Aerial Images Using CNN and Regularization

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Abstract—This study proposes an automatic building footprint extraction framework that consists of a convolutional neural network (CNN)-based segmentation and an empirical polygon regularization that transforms segmentation maps into structured individual building polygons. The framework attempts to replace part of the manual delineation of building footprints that are involved in surveying and mapping field with algorithms. First, we develop a scale robust fully convolutional network (FCN) by introducing multiple scale aggregation of feature pyramids from convolutional layers. Two postprocessing strategies are introduced to refine the segmentation maps from the FCN. The refined segmentation maps are vectorized and polygonized. Then, we propose a polygon regularization algorithm consisting of a coarse and fine adjustment, to translate the initial polygons into structured footprints. Experiments on a large open building data set including 181 000 buildings showed that our algorithm reached a high automation level where at least 50% of individual buildings in the test area could be delineated to replace manual work. Experiments on different data sets demonstrated that our FCN-based segmentation method outperformed several most recent segmentation methods, and our polygon regularization algorithm is robust in challenging situations with different building styles, image resolutions, and even low-quality segmentation.

Index Terms—Building extraction, fully convolutional network (FCN), polygon regularization, segmentation.

I. INTRODUCTION

THE study of automatic building extraction from remote-sensing images not only has important engineering applications such as topographic mapping and household surveys but also has theoretical significance: highly structured, closely human-related characteristics of buildings play a representative role for assessing statistics and machine-learning algorithms and their progress on image segmentation, objection detection, regularization, etc. In the past decades, conventional methods tried to recognize buildings by texture, line, shadow, and more sophisticated, empirically designed features, but failed in automation of building extraction, and no commercial

software has appeared yet. The recent development of deep-learning methods has promoted a new round of research studies toward automation. However, many key questions, e.g., the requirement of large and accurately annotated data sets, the gap between pixel-based segmentation and structured footprints, are still left open. The study of building extraction could be classified into three levels: pixelwise segmentation, buildingwise segmentation, and structured building footprint delineation. In previous works [1], [2], we have shown highly accurate pixelwise and buildingwise segmentation being achieved by using the convolutional neural network (CNN)-based methods on a proposed open and large remote-sensing building data set. In this article, we focus on the ultimate level: building footprint delineation. The objective of this level aims at accurately and automatically delineating structured building boundaries from remote-sensing imagery to replace manual works that has to be fully employed in the past and present, in the surveying, mapping, and GIS applications. In this article, we propose and evaluate a novel building polygon extraction framework integrating the representation learning ability of CNN and the insight of empirical designing and discuss how close we are to a fully automatic building footprint delineation.

Applying deep-learning techniques to pixel-based building segmentation has been well studied [3]–[5] and proved to be more successful than the conventional semantic segmentation methods [6], [7]. The principle behind it is the representation of learning ability through multilevel abstractions of neuron layers that have significantly gone beyond traditional manual feature engineering. The fully convolutional networks (FCNs) are the most popular CNN structures in building segmentation [8] where a binary map of buildings can be efficiently produced with the same size as the original input. Specific FCN structures are developed for building extraction and strategies are designed for the properties of buildings. For example, in order to improve the extraction performance of dense buildings, signed-distance labels [9] were used to provide distances between buildings. A joint distribution of input–output variable was modeled to predict and rectify the replacements for major segmentation errors [10]. Scale effects [1], [2], [4], [11] and radiation differences [1], [12] are the typical problems from using remote-sensing images instead of natural images, which have been actively investigated and are still in the development stage for more robust and accurate segmentation.

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Buildingwise segmentation is commonly categorized into an object-based detection task. Individual buildings and the number of buildings could be retrieved via bounding boxes of each building, while in an original pixel-based segmentation a predicted map only separates pixels instead of buildings. Region-based CNNs (R-CNNs) are currently the representative methods [13]. The Faster R-CNN [14], SPP [15], and YoLo [16] are designed to detect the bounding boxes of objects; the Mask R-CNN [17] is designed to extract bounding boxes as well as pixels of an object in each box. Mask R-CNN has been applied to building instance segmentation [1], [18]. Although the Mask R-CNN could obtain object-level segmentation, it is observed that the boundary of a building is over-smooth and less accurate compared to an FCN-based segmentation. This is because the Mask R-CNN loses details when directly up-sampling very small feature maps to the full size of the input image.

Structured building footprint delineation from remote-sensing imagery has rarely been studied. This is due to the fact that a previous level, accurate pixelwise segmentation, has not been achieved by conventional methods. Hence, the irregular and noisy boundaries are impossible to be regularized to a structured building footprint as manual delineation. The conventional studies of delineating building polygons must introduce more data sources, such as the digital surface model [19]–[21] and LiDAR cloud points [22], [23]. The only deep learning-based building polygon extraction study [18] used the Mask R-CNN for building segmentation and a regularization algorithm to polygonize segmentation results. However, the Mask R-CNN does not accurately segment at the boundary, which is critical for the polygon regularization. The regularization algorithm is not detailed and hard to follow and reproduce. In addition, the experiments were executed on low ground resolution satellite images with inaccurate labels, causing the ability of the current algorithm largely underestimated.

In this article, we propose an FCN-based structure for building segmentation and an empirical polygon regularization for structured footprint extraction. We introduce a multiscale aggregation in an FCN and two postprocessing strategies for achieving very accurate binary segmentation. We propose a specific, robust, and effective polygon regularization algorithm consisting of a coarse and fine adjustment that transforms the segmented building boundary to structured footprints. We assess our building polygon extraction algorithm on a large open building data set [1] and proved more than half of the buildings delineated by the algorithm reaching the manual delineation. To the best of our knowledge, this is the first time such a high level of automation is reported.

The building polygon delineation algorithm consisting the FCN-based segmentation and the empirical regularization is described in Section II. The experiments on different data sets are described in Section III. Discussion and conclusion are given in Sections IV and V, respectively.

II. METHODS

The building footprint extraction algorithm includes pixel-based segmentation, vectorizing and polygonization, and

regularization. First, we develop a multiscale aggregation FCN (MA-FCN) to extract building pixels and introduce two postprocessing strategies to refine the segmentation results. Then, the building pixels are vectorized to irregular boundaries and thereafter polygonized by the Douglas–Peucker algorithm [24]. Finally, the initial polygons are processed by our regularization algorithm to produce structured building footprints.

A. MA-FCN

We develop an MA-FCN architecture for building segmentation, which focuses on the scale effects of remote-sensing data. A feature pyramid network (FPN) [25]-based structure is utilized as the backbone of the MA-FCN, which has a four-layer VGG-16 encoder and a corresponding decoder with lateral connections between them (Fig. 1). Although the FPN structure contains multiscale feature information, it does not make full use of them but only uses the feature maps at the bottom layer (the highest spatial resolution layer) for prediction. As the scale information is critical in remote-sensing data (in this study, the effects of different sizes of buildings in building extraction), we introduce a feature fusion structure to FPN. We implement multiscale feature aggregation by concatenating the last convolution layers of each scale in decoding and predict the final result by using these concatenated feature maps to achieve the comprehensive utilization of multiscale features. These layers (red in Fig. 1) are up-sampled to the original resolution, concatenated, and operated with a 1×1 convolution to obtain the final prediction. In training, the other four predictions of different scales share the same weight with the final prediction and help train scale-dependent network parameters. In the test, only the final prediction embedded with aggregated multiscale representations is used for the network’s output and accuracy assessment. The following equation shows the loss function we used in the training stage:

$$\text{Loss} = \sum_{i=1}^{n+1} L_i \quad (1)$$

where $n+1$ represents four predictions at each scale (1/1, 1/2, 1/4, and 1/8) plus the final prediction (see Fig. 1). Each loss L_i is computed utilizing a cross-entropy loss.

We also introduce batch normalization at convolutions of each scale of the encoder for accelerating convergence and preventing overfitting.

Our MA-FCN structure is more compact and robust than other FCN-based building extraction methods. For example, in a very recent study [2], we found introducing a set of Atrous convolution [26] and ASPP [27] led to unstable experimental results; in the work of [11], the multiscale information was implemented only with averaging outputs of different scales of an FPN, which led to lower scale robustness.

Our segmentation algorithm additionally includes two simple but effective postprocessing strategies to improve segmentation results. The first one is to produce an optimal building map by fusing models trained separately. For each pixel, only when the voting from most of the models is consistent,

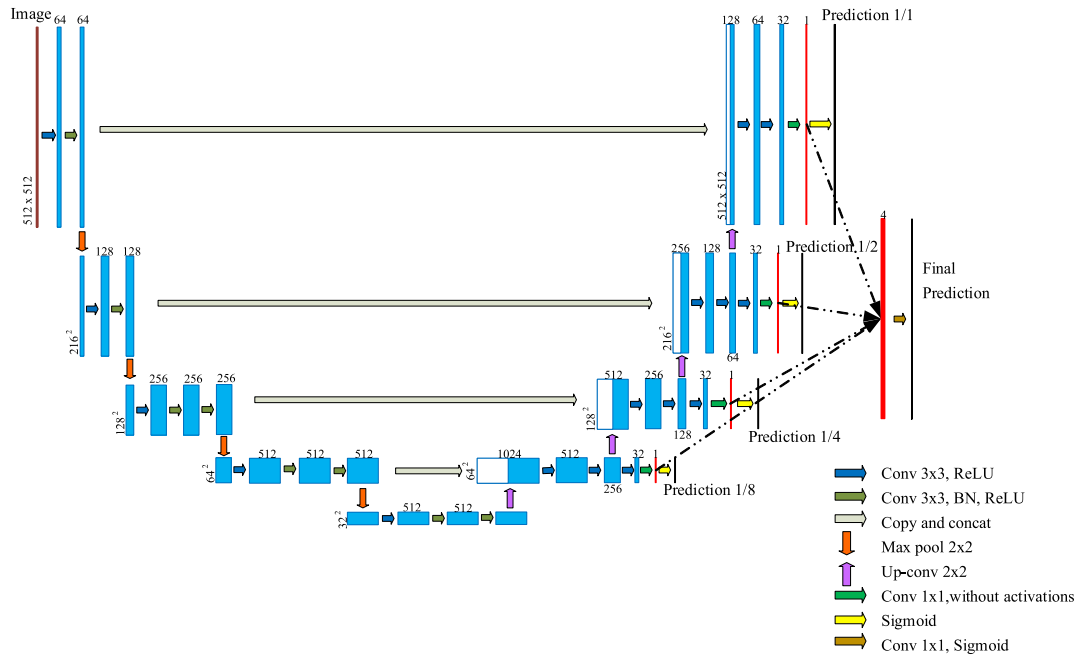


Fig. 1. MA-FCN structure.

it is labeled as building or nonbuilding. In practice, we train three models and check if more than two models predict a probability over or below a given threshold (0.5 as a default setting in a binary segmentation). This strategy can improve the intersection over union (IoU) score 0.6% from already very high IoU (about 90%) on the WHU aerial data set [1].

The other strategy is to fuse the prediction maps in the overlapped area to build a seamless segmentation map of the whole test area. Owing to the limitation of GPU capacity, large-size remote-sensing images have to be cropped into image tiles (e.g., 512×512 pixels) to feed a CNN. The problem associated with this is the boundary effect, that is, the boundaries of each tile are more likely to be incorrectly labeled especially when applying a convolution where padding operation must be applied to the edge of an input tile. Although this boundary effect may have little impact on the accuracy assessment of a binary segmentation [1], it does affect a building polygon regularization task as the border of a building needs to be accurately delineated. Fig. 2(a) shows an example of such a problem, where the lower right building was split into different tiles and cannot be restored into a whole building because of the prediction errors in the lower right tile [Fig. 2(b)]. In order to address this problem, we applied an overlapping and cutting strategy to avoid using the prediction results of tile boundary pixels and produced a more favorable map as shown in Fig. 2(c). First, we predict the image of the test area on 50% overlapping tiles rather than on seamless cropped tiles. Then, we cut off the 25% border of each tile (i.e., avoid the use of predictions of border pixels) and seamlessly stitch all of these tiles to the whole image of the test area. This strategy is simple, but it provides an effective solution to the boundary effect in CNN-based remote-sensing applications. This preprocessing gained 0.6% IoU improvement on the WHU data set.

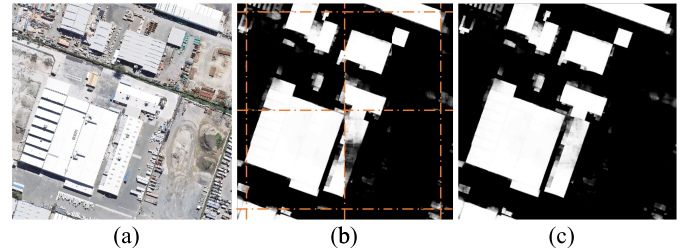


Fig. 2. (a) Original image. (b) Boundary effects caused by the seamlessly cropped tiles. (c) After the overlapping and cutting strategy is applied.

B. Boundary Regularization

The binary segmentation map produced by the MA-FCN is preprocessed by a boundary extraction and a polygonization. Each connected area consisting of building pixels is grouped into an individual building, and the boundary extraction is implemented with the Marching Cubes algorithm [28]. The polygonization is then implemented with the Douglas–Peucker algorithm [24].

Starting from the initial polygons, our boundary regularization algorithm consists of two steps: a coarse adjustment that eliminates obvious errors from segmentation and polygonization and a refinement that adjusts line’s directions and node’s position.

Owing to the imperfect segmentation maps and the greedy Douglas–Peucker algorithm of polygonization, the initial polygons exhibit many errors (Fig. 3), which causes a serious problem for footprint extraction. To reduce this kind of errors, we introduce an algorithm (Algorithm 1), which consists of empirical rules.

Algorithm 1 uses simple but effective rules excluding those most impossible nodes, lines, and polygons. For example, a building edge should have at least a certain length, and

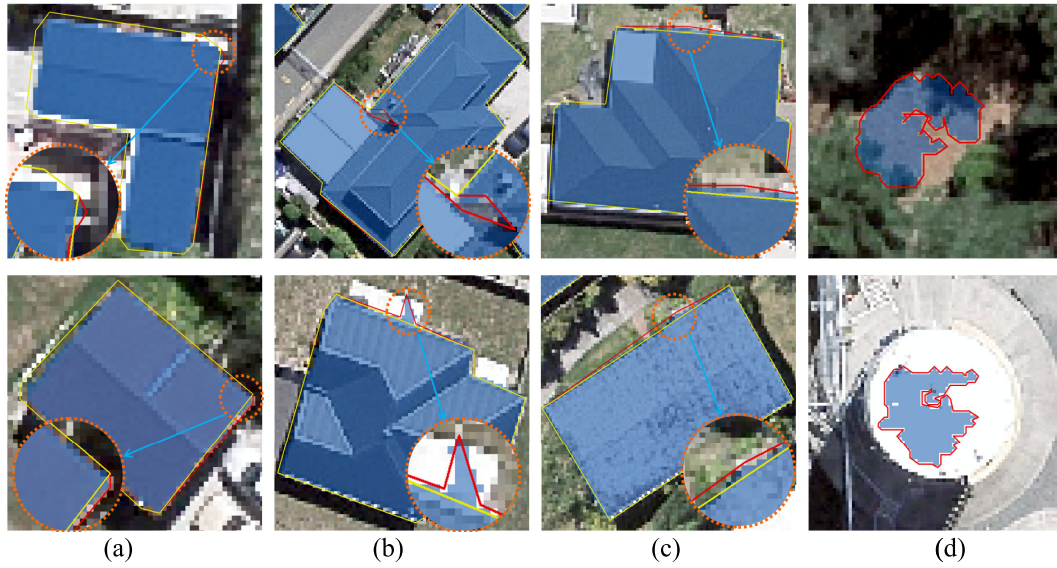


Fig. 3. Examples of coarse adjustment where original red lines are replaced with yellow lines or removed. (a) Short edge elimination. (b) Sharp angle elimination. (c) Smooth angle elimination. (d) False building polygon removing.

Algorithm 1 Coarse Adjustment

1. Remove polygons whose area is below a threshold S .
 2. Remove edges whose lengths are below a given side length T_d that varies with the area of a building.
 3. Remove over-sharp angles with threshold α .
 4. Remove over-smooth angles with threshold β .
-

we set a threshold value T_d that changes with the area of current building candidate. In Fig. 3, red lines are the polygons extracted by the Douglas–Peucker algorithm. By excluding the short lines in Fig. 3(a), polygons are refined and adjusted to better visual representations (yellow line) for further processing. These rules also result in the elimination of obvious false polygons as Fig. 3(d). The angle thresholds α and β exclude nodes resulting in over-sharp or over-smooth angles, as shown in Fig. 3(b) and (c).

Our fine polygon regularization algorithm (Algorithm 2) is inspired by Awrangjeb [22] where a regularization algorithm is introduced for extracting building polygons from point clouds. Our algorithm leverages the idea of main directions, which is also widely used in multiple tasks in vision processing [29], [30].

Algorithm 2 lists the pipeline of our fine regularization. In the first step, we set different edge thresholds W according to a certain building area. Edges longer than W are treated as long edges, otherwise short edges. For smaller buildings, we gradually reduce W until at least one long edge is located. The direction of the longest edge [e.g., the left green line in Fig. 4(c)] is then added to a vacant list as the initial main direction. We also consider the situations where more than one main direction exists in a complex building structure in step 3: all the other long edges are compared with the main directions in the current list. If the angle between a long edge and any main direction falls into the range $[\delta, \pi/2 - \delta]$, the edge's direction is added to the list. In step 4, a long edge will be rotated to the main direction (or its perpendicular direction)

Algorithm 2 Fine Adjustment

1. Find long edges with threshold W that varies with building's area.
 2. Add the direction of the longest edge to the list of main direction.
 3. Add other edges' direction to the list of main directions according to the angle threshold δ between their directions and directions in the list.
 4. Adjust long edges according to the list and angles.
 5. Adjust short edges (judged by a threshold θ) according to the list and angles.
 6. Merge (or connect) parallel lines if the distance between two lines is less than (or larger than) a threshold d .
 7. Connect all the adjusted lines to form the final polygon.
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around the midpoint of its vertices if the angle between it and the main direction (or its perpendicular direction) is less than δ or larger than $\pi/2 - \delta$ [see Fig. 4(d)]. In step 5, threshold θ is set to adjust all the short lines to be parallel or perpendicular to the main directions [see Fig. 4(d)]. If the angle between a short line and the closest main direction is less than θ , this short line is rotated around the midpoint to be parallel to the main direction; otherwise, the line is rotated to perpendicular to the main direction. In step 6, the distance threshold d between parallel lines is set according to the area of the building. For example, in region 1 of Fig. 4(e), a short perpendicular line is added to connect the parallels; in region 2, a shorter line in the parallels is removed, both by comparing with the threshold d . The final step calculates the intersection of the adjacent edges (or their extension lines) in order and then sequentially connects the nodes to produce the final polygon [Fig. 4(f)].

III. EXPERIMENT AND RESULTS

A. Data Set and Settings

We applied our algorithms to the WHU aerial building data set [1] for evaluation. In Fig. 5, the data set covers

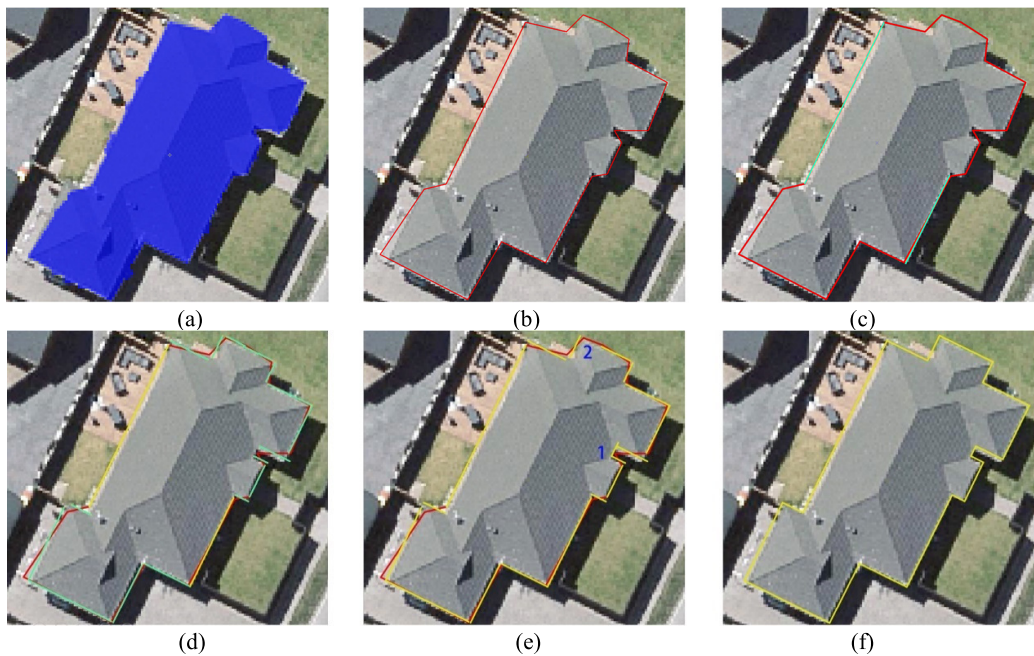


Fig. 4. Procedure for fine adjustment. (c) Long edge adjustment. (d) Short edge adjustment. (e) Parallel lines merging in area 2 and parallel lines connecting with a new line in area 1. (f) Final polygon. (a) and (b) Segmentation result and polygonization result, respectively.



Fig. 5. Aerial data set in Christchurch, New Zealand, covering 450 km² with 0.2 ground resolution (cited from [1]). The areas in the blue, yellow, and red bounding boxes are used for training, validation, and test, respectively.

181 000 buildings of various usages, architectures, and colors. Among them, 130 000 buildings in the blue box are used for training, 14 500 buildings in the yellow box are used for validation, and the rest 36 700 buildings in the two red boxes are used for testing. The original resolution of the aerial data sets is 0.75 m. We down-sampled the data sets to 0.2 m. To fit into the GPU capacity, the whole image is cropped into 512×512 tiles.

Our algorithm is implemented with the Keras platform using a TensorFlow backend on Windows 10 system. In training, five 512×512 image tiles consist of a mini-batch and the Adam (adaptive moment estimation) algorithm is utilized for optimization with a learning rate 0.0001, β_1 0.9, β_2 0.999, and ε $1e-8$. Except for the last layer of 1×1 convolution is initialized with weight 0.25, the weights of filters are initialized with a normal distribution and the bias with zeros.

TABLE I
EXPERIMENTAL RESULTS OF DIFFERENT SEGMENTATION METHODS
APPLYING TO THE WHU AERIAL DATA SET

Method	IoU	Recall	Precision
Si-Unet [1]	0.886	0.848	0.931
U-Net [31]	0.879	0.943	0.929
Deeplabv3+ [32]	0.878	0.935	0.935
Mask R-CNN [17]	0.849	0.823	0.914
MA-FCN	0.895	0.942	0.945
MA-FCN w/ overlap	0.901	0.948	0.949
MA-FCN w/ overlap & vote	0.907	0.951	0.952

We used an Nvidia 1080 TI 11 G GPU. The training of the aerial data took 7 h. All the output probabilities of each pixel from a sigmoid classifier are translated to binary values with a threshold of 0.5. Three representative indicators, IoU, precision, and recall are used for evaluation and comparison of different methods.

B. Building Segmentation

Table I shows the segmentation results of our method, recent benchmark methods for general semantic segmentation, U-Net [31], Deeplabv3+ [32] and Mask R-CNN [17], and a method especially for building extraction, Si-Unet [1], on the test area (36 700 buildings). Our MA-FCN outperforms the second-best method Si-Unet 0.9% on the IoU. When the two postprocessing strategies are added, our method reaches 0.907 IoU and is 2.1% and 5.8% higher than the Si-Unet and Mask R-CNN. As both recall and precision have reached a very high score (about 95%), the 2.1% IoU improvement is a considerable achievement. More importantly, the high accurate segmentation constitutes the key requirement for the automation of building footprint extraction.

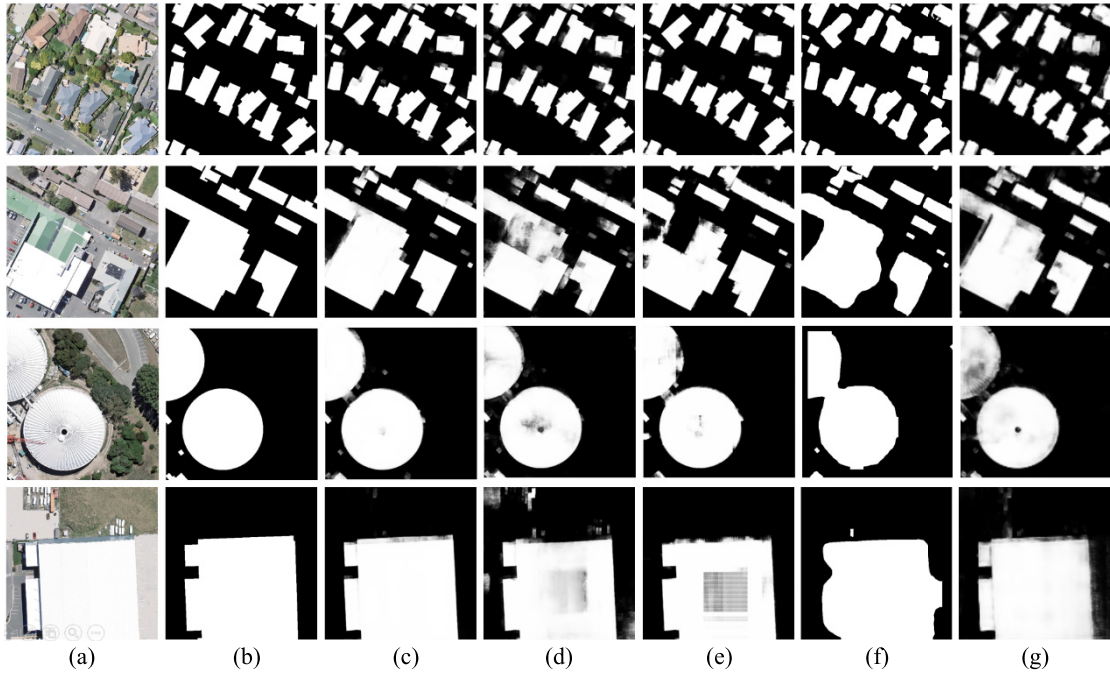


Fig. 6. Segmentation results of different methods on the WHU data set. (a) Images. (b) Label. (c) Our MA-FCN without postprocessing. (d) Si-Unet. (e) U-Net. (f) Mask R-CNN. (g) Deeplabv3+.

Although the current benchmark segmentation method Deeplabv3+ introduced the Atrous convolution in the encoder, it does not show improvement on the aerial building data set compared to the simpler U-Net. The Si-Unet outperformed these two methods with about 1% rise due to the introduction of two-scale inputs, i.e., inputs with full resolution and half resolution. This strategy partially solved the multiscale effects in remote-sensing data but doubled both sample capacity and training time. Our MA-FCN used a more dedicated way to fully utilize the multiscale information from the existing feature pyramids that are abstracted via convolution layers, without requirements of additional computation and more memory capacity. The Mask R-CNN functions the worst due to its mechanism that only a very shallow FCN is used for segmentation, where the direct up-sampling from very small feature maps to the prediction map of original size caused loss of details, especially at the objects' boundaries.

Fig. 6 shows four examples of segmentation results of different methods on the 512×512 tiles. The segmentation map produced using our MA-FCN [Fig. 6(c)] is the most similar to the label [Fig. 6(b)]. The Si-Unet [Fig. 6(d)] and U-Net [Fig. 6(e)] could not predict precisely in the middle part of buildings in the last two rows of Fig. 6 as they do not fully utilize the multi-scale information. The Mask R-CNN [Fig. 6(f)] only outputs binary maps where the building boundary is obviously over-smooth and is worse than the FCN-based methods. All the predicted maps from the Deeplabv3+ [Fig. 6(g)] also show to be less accurate than our MA-FCN predictions.

C. Building Polygon Regularization

Based on the areas of buildings, we classify them into ordinary buildings ($<200 \text{ m}^2$) and large buildings ($\geq 200 \text{ m}^2$). We set the parameters related to the two building types,

i.e., the tolerance distance D in the Douglas–Peucker algorithm, the minimum side length T_d in the coarse adjustment (Algorithm 1) and the minimum length of a long edge W in the fine adjustment (Algorithm 2), as

$$\begin{cases} D = 0.3, & T_d = 0.5, & W = 6; & S \leq 200 \\ D = 1.0, & T_d = 0.9, & W = 8; & S > 200. \end{cases} \quad (2)$$

The average lengths of ordinary and large buildings in the WHU data set are 6.4 and 7.9 m, respectively; we set W to 6 and 8, respectively, to discriminate long and short sides. Analogously, T_d is set according to the building area to eliminate redundant nodes. Another threshold d discriminating merging or connecting parallel lines in Algorithm 2 is highly related to T_d . We set $d = T_d + 0.3$.

The other parameters in the coarse adjustment that aim at removing those unreliable candidate buildings and nodes are explicit. We set $\alpha = \pi/6$ and $\beta = 17\pi/18$ to remove nodes resulting in too sharp and too smooth angles, respectively, and set $S = 20 \text{ m}^2$ to exclude too small building candidates.

The other two parameters in the fine adjustment aim at adjusting edges to form regular building polygons, which is explicit and applicable to most types of buildings. We set $\delta = \pi/12$ to determine whether to add the direction of a long side to the main direction list. This parameter guarantees those buildings with nonorthogonal polygon shapes accurately delineated. The short sides are adjusted to be parallel or perpendicular to the main direction according to $\theta = \pi/4$. The adjustment of the short sides has a much smaller effect on the polygon than that of the long sides.

Although there are a few parameters seeming to be set empirically, the following experiments in different data sets of different resolution and on segmentation maps of different

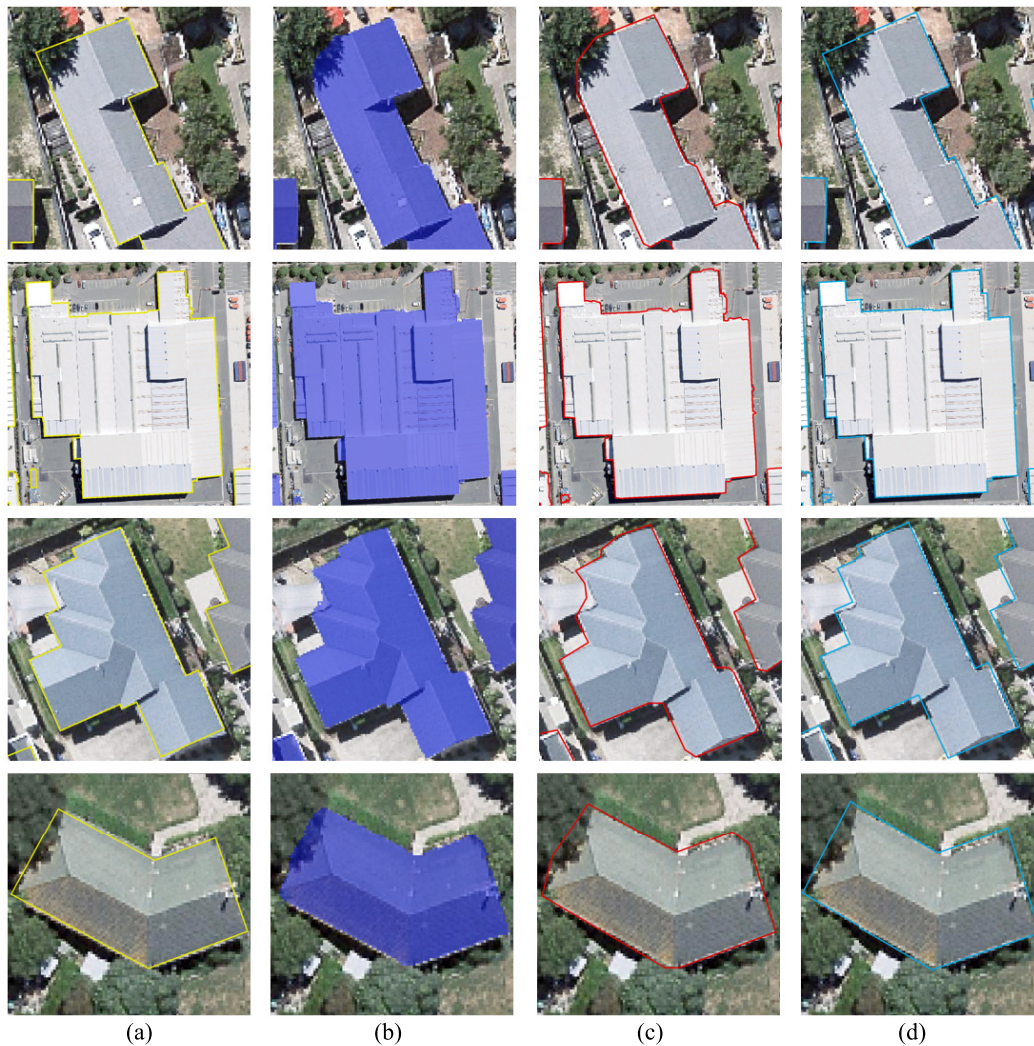


Fig. 7. Examples of our polygon regularization results. (a) Images with labels. (b) Segmentation results of our MA-FCN. (c) Initial polygon from the Douglas–Peucker algorithm. (d) Results of applying our polygon regularization algorithm.

accuracy all show that our regularization algorithm with these fixed threshold settings is robust.

Fig. 7 shows four regularization results on the WHU aerial data set where building footprints were accurately extracted. In the first row, the shades from a tree that covered a corner of the building hamper a successful segmentation. By adjusting the two lines at the corner according to the main direction (the longest line), the two lines intersect at the correct node. The second example shows that the algorithm is robust in situations of large and complex buildings with many corners. The building in the third row of Fig. 7 shows a type of building that is common in this area, the (b) and (c) of the figure shows an error at the upper-left corner. This error is typical due to the imperfect segmentation and polygonization. However, it is well repaired by our regularization algorithm (d). The fourth row of Fig. 7 shows a building that has two main directions (a nonorthogonal polygon case), and our algorithm realized accurate polygon regularization by recording a list of main directions.

A comparison between Fig. 7(a) and (d) shows that our regularization algorithm is capable of delineating building footprints almost as accurate as manual annotation, despite

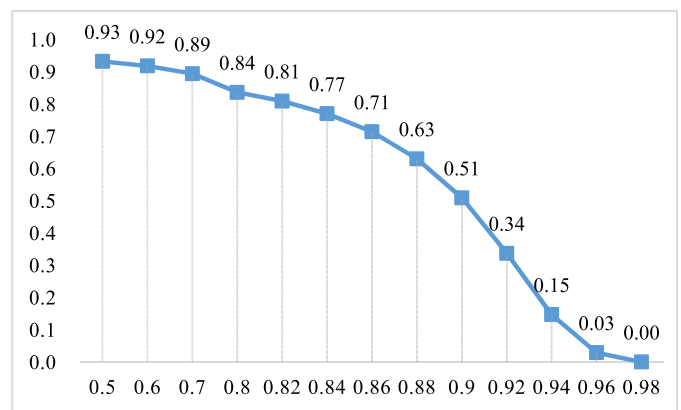


Fig. 8. Percentage of valid polygons with respect to the different IoU thresholds on the WHU data set using our algorithm.

the obvious errors from the segmentation and polygonization steps.

Table II shows the accuracy changes after initial polygonization and polygon regularization. Using the classic Douglas–Peucker algorithm, the transformation from the



Fig. 9. Large area covered with valid polygons (blue) and those polygons with less than 90% IoU (red). However, most of the red polygons show no difference to a manually delineated footprint, indicating our threshold is very rigorous.

TABLE II
ACCURACY OF SEGMENTATION, POLYGONIZATION, AND POLYGON
REGULARIZATION ON THE TEST AREA OF THE WHU DATA SET

Stage	IoU	Recall	Precision
Segmentation	0.907	0.952	0.951
Polygonization	0.895	0.970	0.921
Regularization	0.892	0.966	0.920

binary raster image to vector polygon caused 1.2% drop in the IoU. Our polygon regularization algorithm functioned on the initial polygons causes almost no accuracy loss.

Finally, we plot a trend chart that reflects the change of valid building polygon number with respect to the IoU (Fig. 8). A valid building polygon means the polygon does not require further manual adjustment or delineation and can be used directly as a final footprint product.

We find IoU is an effective indicator for this discrimination. Suppose an ideal building with the size 100×100 pixels ($20 \text{ m} \times 20 \text{ m}$ here), and the manually delineated polygon contains 2–3 pixels bias in each edge. If the square is shifted 2 pixels in both horizontal and vertical directions, the IoU would be 92.4%, and if 3 pixels, the IoU would be 88.8%. Hence, we set the threshold of IoU to 0.9. If an automatically extracted polygon satisfies $\text{IoU} > 0.9$, we treat it as a valid footprint. According to Fig. 8, 51% of buildings (18 691 out of 36 700) satisfy this condition. It means more than half of manual works could be replaced.

The threshold setting is also well justified by visual checking. We superimposed those valid polygons on the images. In Fig. 9, all the valid polygons (blue) show almost no difference from a manual delineation. It is observed that most

of those invalid polygons (red) also show no difference to manually delineated footprints, indicating our threshold is very rigorous. According to Fig. 8, when the threshold loosens to 0.88 and 0.86, there are exist more than 60% and 70% valid polygons, respectively. In fact, with a threshold of 0.86, we found that all the valid polygons are accurately delineated by visual checking. The results indicate our algorithm is promising toward automatic building polygon extraction.

D. More Data Sets

We conducted a further evaluation of our algorithm with other data sets. Aerial images covering Tyrol (Switzerland), Kitsap (Washington, DC, USA), and Austin (TX, USA) are selected from the Inria data set [12]. In each city, there are $36\,5000 \times 5000$ aerial images with 0.3-m ground resolution. We classified the training area and test area with 3:1. All the parameters are kept the same as in the WHU data set. Table III shows the results of the MA-FCN segmentation and regularization of our method. The average IoU score dropped 13% compared to the WHU data set, this is because fewer training samples and more labeling errors exist in the Inria data set [12]. Compared to U-Net, Si-Unet, Deeplabv3+, and Mask R-CNN, our MA-FCN method performed substantially better in all the three cities. The overlapping and cutting strategy increased the IoU of the MA-FCN method by 1%. It can be observed that with the regularization algorithm, the IoU decreased trivially even when the segmentation results are poor.

Fig. 10 shows some examples of segmentation and regularization results of our algorithm. It is observed that the boundaries of buildings (red) are less accurate compared to

TABLE III
ACCURACY OF DIFFERENT SEGMENTATION METHODS ON THE INRIA DATA SET

City	Tyrol			Kitsap			Austin		
Method	IoU	Recall	Precision	IoU	Recall	Precision	IoU	Recall	Precision
U-Net [31]	0.713	0.828	0.837	0.670	0.741	0.876	0.766	0.858	0.877
Si-Unet [1]	0.714	0.803	0.866	0.682	0.842	0.782	0.768	0.865	0.873
Deeplabv3+ [32]	0.705	0.827	0.828	0.665	0.734	0.826	0.756	0.821	0.905
Mask R-CNN [17]	0.744	0.852	0.854	0.661	0.738	0.863	0.767	0.858	0.879
MA-FCN	0.769	0.870	0.869	0.718	0.812	0.861	0.803	0.885	0.897
MA-FCN w/overlap	0.779	0.878	0.873	0.728	0.818	0.869	0.812	0.893	0.899
Regularized polygon with MA-FCN	0.775	0.904	0.847	0.721	0.866	0.811	0.795	0.911	0.861

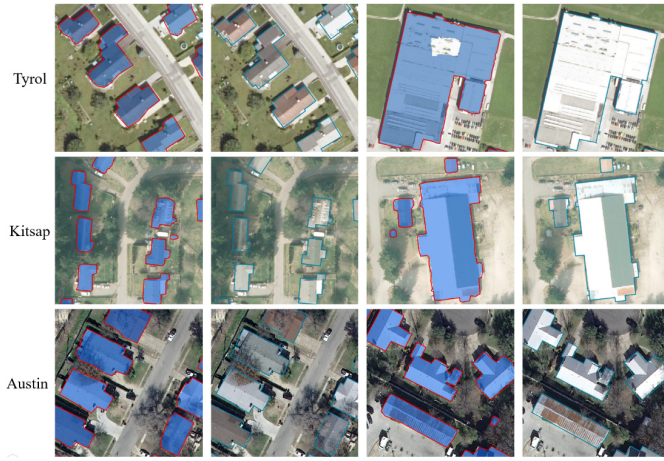


Fig. 10. Our polygon regularization algorithm on the Inria data set.

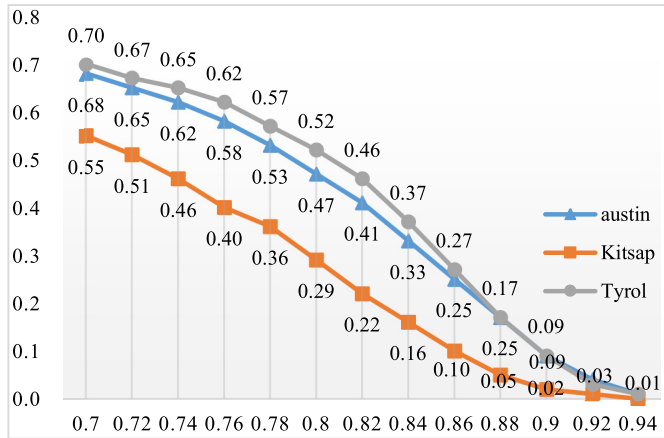


Fig. 11. Percentage of valid polygons with respect to the different IoU thresholds on the Inria data set with our algorithm.

manual delineation. However, our regularization algorithm revised all the ragged edges and guaranteed the building footprint delineation (blue) approaching manual delineation. It also demonstrates that the parameters of the regularization algorithm are robust to different data sets with different building styles and resolutions.

We count the valid building proportion with respect to the IoU as in the WHU data set. The curves in Fig. 11 indicate only 9% of buildings in Tyrol and Austin could be classified as valid if using 0.9 as the threshold. It is even worse in Kitsap



Fig. 12. Large area of Austin covered with valid polygons (blue) and polygons with less than 86% IoU (red). However, many polygons in red also show as accurate as a manually delineated footprint.

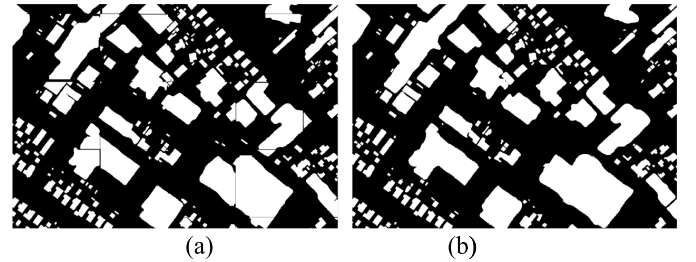


Fig. 13. Segmentation results of the Mask R-CNN (a) before and (b) after using the overlapping and cutting strategy.

city. By comparing the results of the WHU data set where 51% of polygon footprints are considered valid, we see the automation level of building footprint extraction relies strongly on the previous segmentation. It implies in the current stage large and accurate annotated building data set is necessary to structured building polygon extraction.

When we set the threshold to 0.86, the number of valid buildings in the three cities significantly improved. In Tyrol, the percentage of valid building footprints becomes 27%. Fig. 12 shows one large area of Austin where the valid buildings are colored in blue and the rest red. Visually, the valid polygons (blue) are all drawn accurately as manual delineation, and those invalid polygons consist of polygons with both obvious errors and accurately delineated. Therefore, we concluded that 86% may be a good threshold to judge

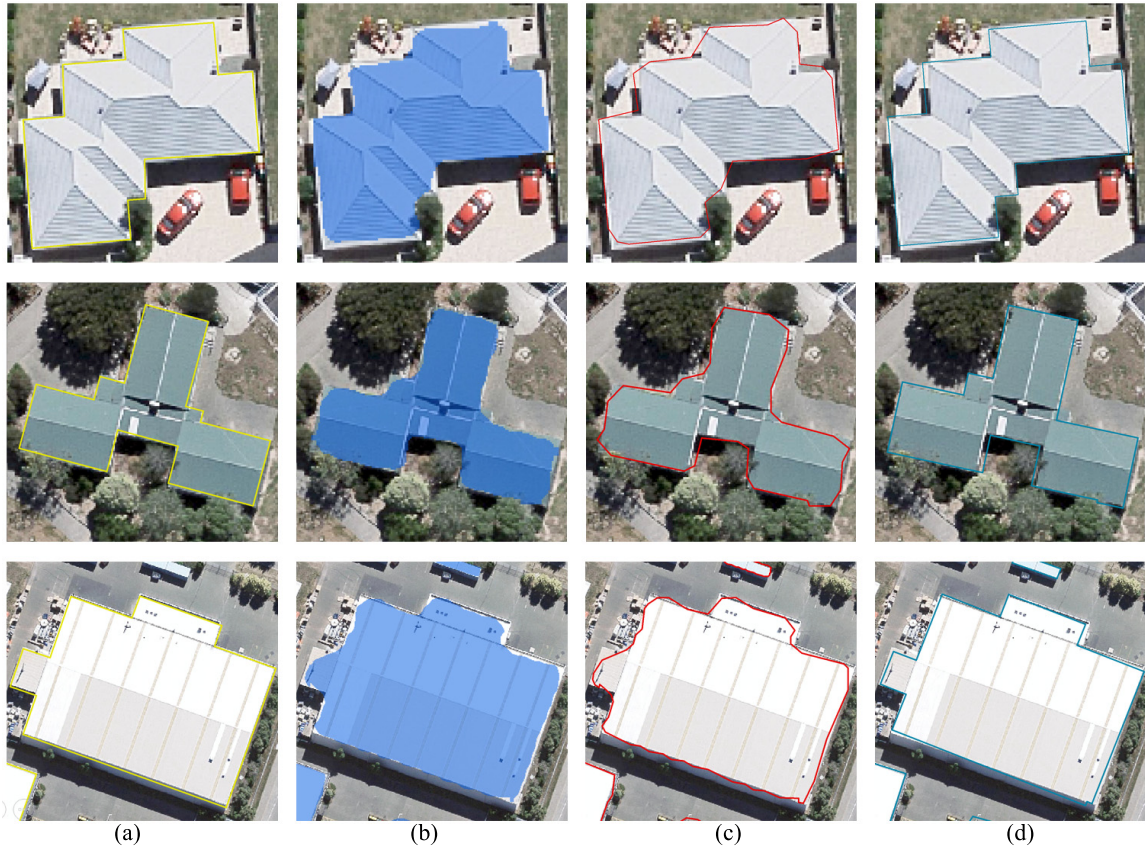


Fig. 14. (d) Examples of our polygon regularization on (b) segmentation maps of the Mask R-CNN. (a) and (c) Labels and initial polygons obtained from the Douglas–Peucker algorithm.

whether a building footprint by the algorithm needs further manual revision.

E. Using Mask R-CNN for Segmentation

A regularization process was used in [18] for only orthogonal-polygon buildings but is not adequately described. The lacking of reproducibility leads to the failure of the comparison of it with our work. Since the Mask R-CNN is used for segmentation in [18], it is also the state-of-the-art instance segmentation algorithm; we further investigate our regularization algorithm on the segmentation results of Mask R-CNN on the WHU data set. The training time is 17 h at the same training data set. Fig. 13 shows that the split buildings in the image tile border are hardly repaired because, in a convolution, padding must be applied to the edge of an input image; however, it could be well compensated by our simple overlapping and cutting strategy. Table IV shows after using our strategy, the IoU considerably improved 2.7% and reach almost the same accuracy of benchmark segmentation methods as the Unet and Deeplabv3+ (see Table I). However, the unfavorable over-smoothing effect of edges is not improved [Fig. 13(b)]. In Table IV, the IoU score decreases very slightly after the polygonization via the Douglas–Peucker algorithm and our boundary regularization, respectively, and shows no obvious accuracy loss.

Fig. 14 also shows the segmentation boundaries drawn by the Mask R-CNN is over-smoothed as it directly scales

TABLE IV
ACCURACY OF THE SEGMENTATION AND REGULARIZATION USING THE MASK R-CNN ON THE WHU DATA SET

Stage	IoU	Recall	Precision
Mask R-CNN	0.849	0.823	0.914
Mask R-CNN & overlap	0.876	0.946	0.921
Polygonization	0.873	0.941	0.923
Regularization	0.869	0.939	0.921

very small features (here 28×28) to the original input size (here 512×512). However, despite the much worse boundaries delineation than that is delineated with an FCN-based method, our regularization algorithm obtained accurate footprints of buildings, which further indicates the robustness and effectiveness of the algorithm in challenging situations.

Fig. 15 shows the changes of valid building numbers with respect to the IoU threshold using the Mask R-CNN as the segmentation algorithm. The 43%, 58%, and 68% building polygons drawn by the algorithm meet the automation level when setting the IoU threshold to 0.9, 0.88, and 0.86, respectively. Although using our MA-FCN for segmentation, the valid building polygon ratio is 51%, 63%, and 71%, respectively. The narrowing gap between their performances indicates that the MA-FCN has a tremendous advantage when the threshold is very rigorous; when the threshold loosens, the Mask R-CNN that segments less accurate boundaries

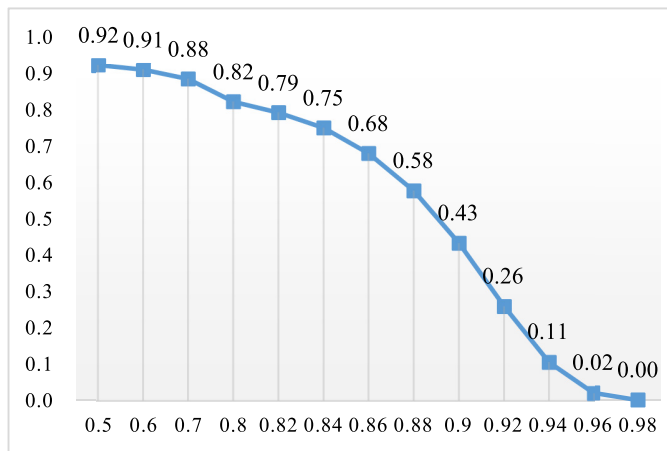


Fig. 15. Percentage of valid polygons with respect to the different IoU thresholds on the WHU data set using the Mask R-CNN (with our overlapping strategy) and our regularization algorithm.

could gradually approach the performance of our footprint extraction algorithm. It further demonstrates that the polygon regularization algorithm is robust to low-quality segmentation.

IV. DISCUSSION

From the performance comparison of our method on the WHU data set and the Inria data set, it is clear that, once the IoU of segmentation reaches 87% (Table IV), we can obtain a high automation level and up to 70% buildings can be delineated by the algorithm; in contrast, when the IoU is below 80% (Table III), the automation level dramatically dropped to below 30%. From this, we could derive an empirical criterion for structured building footprint extraction: the highest possible IoU score of segmentation should be reached, and to at least 87%, before a regularization algorithm like ours should be applied. This IoU score can be achieved by using larger and high-quality training data sets.

Our parameters in coarse and fine polygon adjustments are robust, which guarantees them to be applied directly to the Inria data set. However, the regularization accuracy is sensitive to the change of the tolerance distance D in the Douglas–Peucker algorithm. The initial semantic segmentation map has jagged edges, details will be lost if D is set too big whereas a too small D will make the polygon complex and unrealistic. Altering T_d (the minimum side length) in Algorithm 1 has a similar effect as altering D , but the effect is relatively trivial. Changing the other parameters in Algorithms 1 and 2 affects the accuracy relatively less.

We discuss the effects of the whole coarse adjustment (Algorithm 1). When Algorithm 1 is removed, the final regularization accuracy will drop 0.6% IoU (Table II). This improvement from Algorithm 1 is slight because mainly the boundaries are adjusted and the area of it is very small compared to the mask of the entire building area. However, the simple coarse adjustment can help one to obtain a more natural building structure. Fig. 16 shows two typical examples. In the first row, the ragged edges make it difficult for finding a suitable long side as the main direction, which is of the key

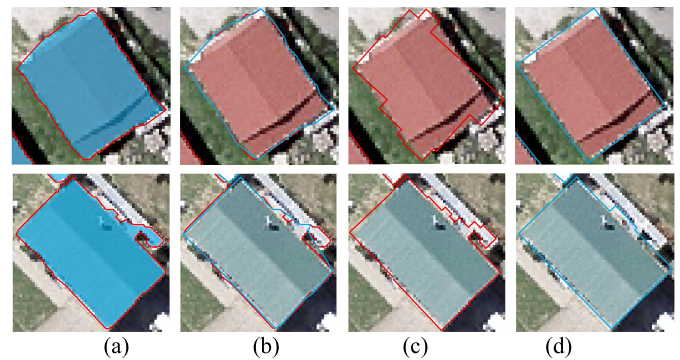


Fig. 16. Examples of the effects of the coarse adjustment. (a) Segmentation results and the initial polygon. (b) Initial polygon (red) and coarse adjusted polygon (blue). (c) Results without coarse adjustment. (d) Results with coarse adjustment.

importance for a successful regularization. Fig. 16(c) shows that obvious errors are caused by applying fine adjustment directly on the initial polygon. The coarse adjustment can eliminate inaccurate short sides so that an ideal long side is obtained at the bottom [Fig. 16(c)] to guarantee an accurate footprint [Fig. 16(d)]. In the second row, the imperfect segmentation map results in many false orthogonal angles. The errors can also be largely eliminated by the coarse adjustment algorithm.

V. CONCLUSION

The end-to-end manner of a CNN has once been considered the way toward automatic AI applications, but recent studies have shown combining CNN and empirical design could be more practical. The building footprint extraction is a typical case where the CNN-based segmentation and empirical designed regularization could bridge the gap between machine recognition and human understanding. This article has shown that on a large high-quality data set, 50%–70% of structured building footprints can be delineated by the algorithm instead of human labor. This high-level of automation has not been reported before. The first part of our algorithm, the CNN-based MA-FCN, has been proven to outperform several most recent methods on different data sets; the second part of our algorithm, the polygon regularization, has shown to be robust to different regions and building styles, image resolutions and even low-quality segmentation. In addition, our algorithm is easily reproducible and can be applied to other data sets and engineering applications.

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